Cost Analysis

2026-08-17

Contents

1 Assumptions

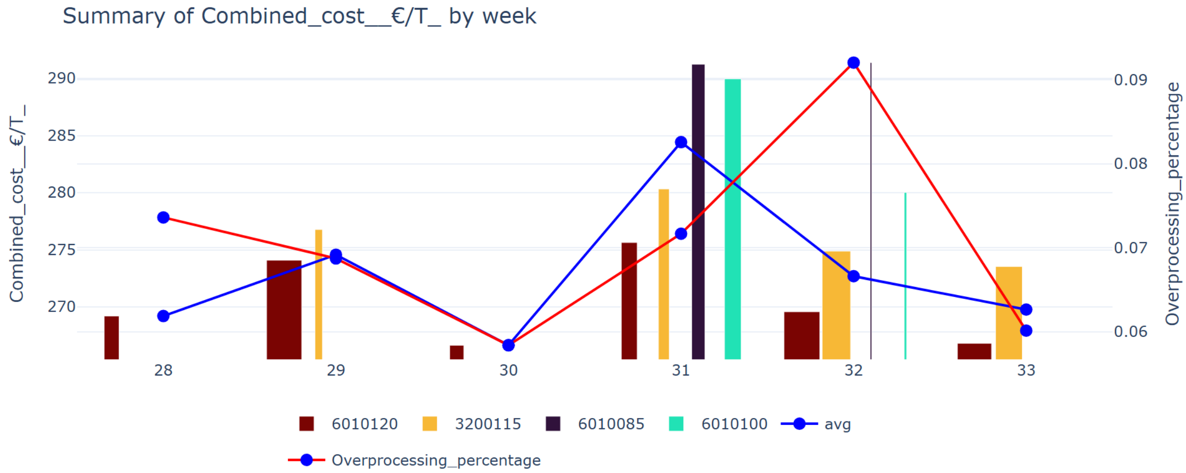
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

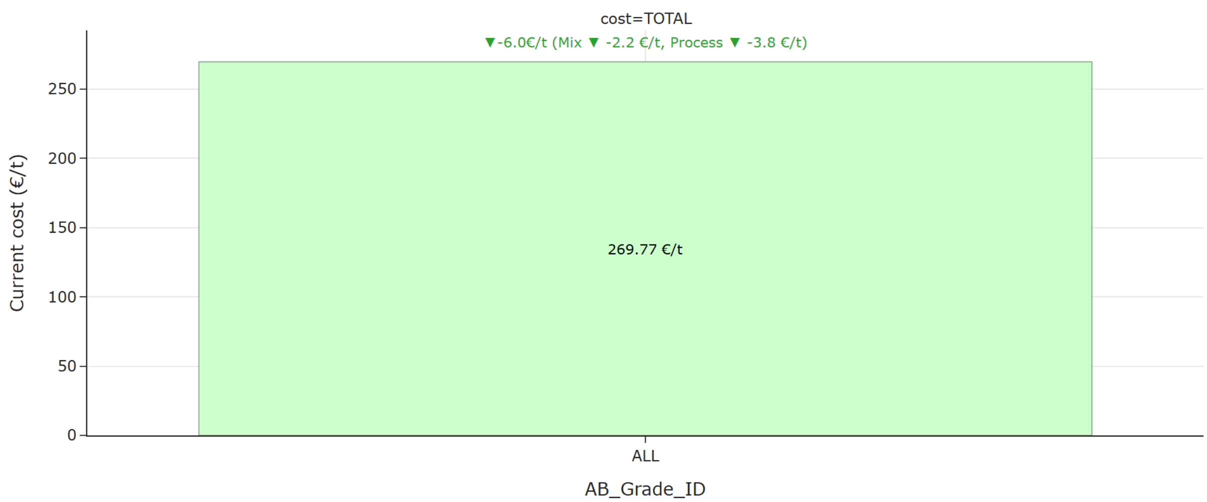
- **Period Analyzed**: July 12, 2026 – August 17, 2026
- **Cost Components Included**: Fibre, steam, electricity, starch, and chemicals.
- **Grades Included**: Only grades 6010085, 6010100, 601020 and 3200115 are included in this week's analysis.
- **Data Selection Criteria**: Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used**:
  - **Current**: Refers to the last week of the analyzed period.
  - **Historic**: Represents the average cost over the monthly interval previous to the "Current" period. Used as baseline.
  - **Best**: Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

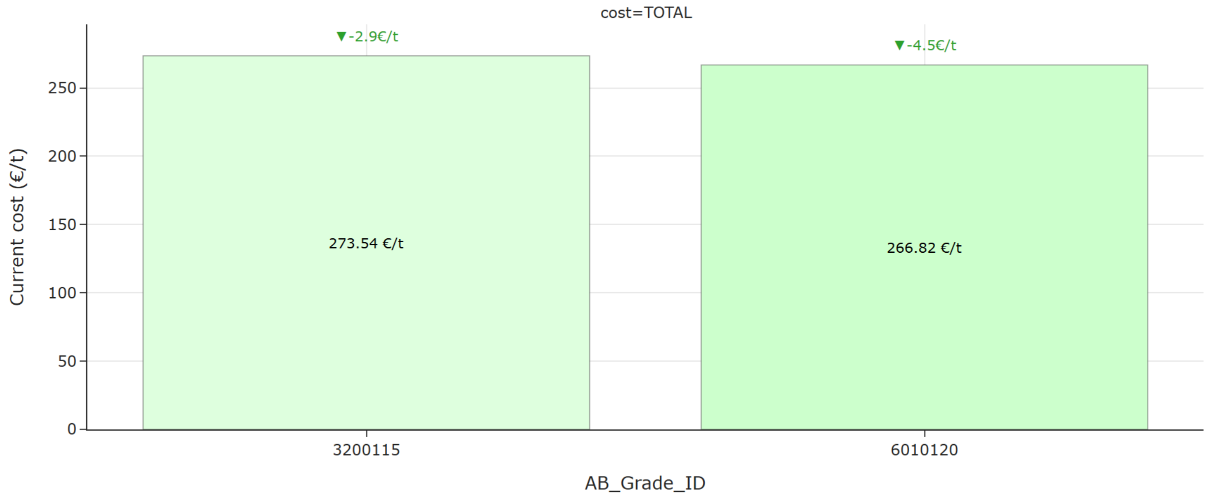
The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.





For all considered grades, TOTAL cost saw a decrease of 6.01 €/t, resulting in 269.77 €/t.

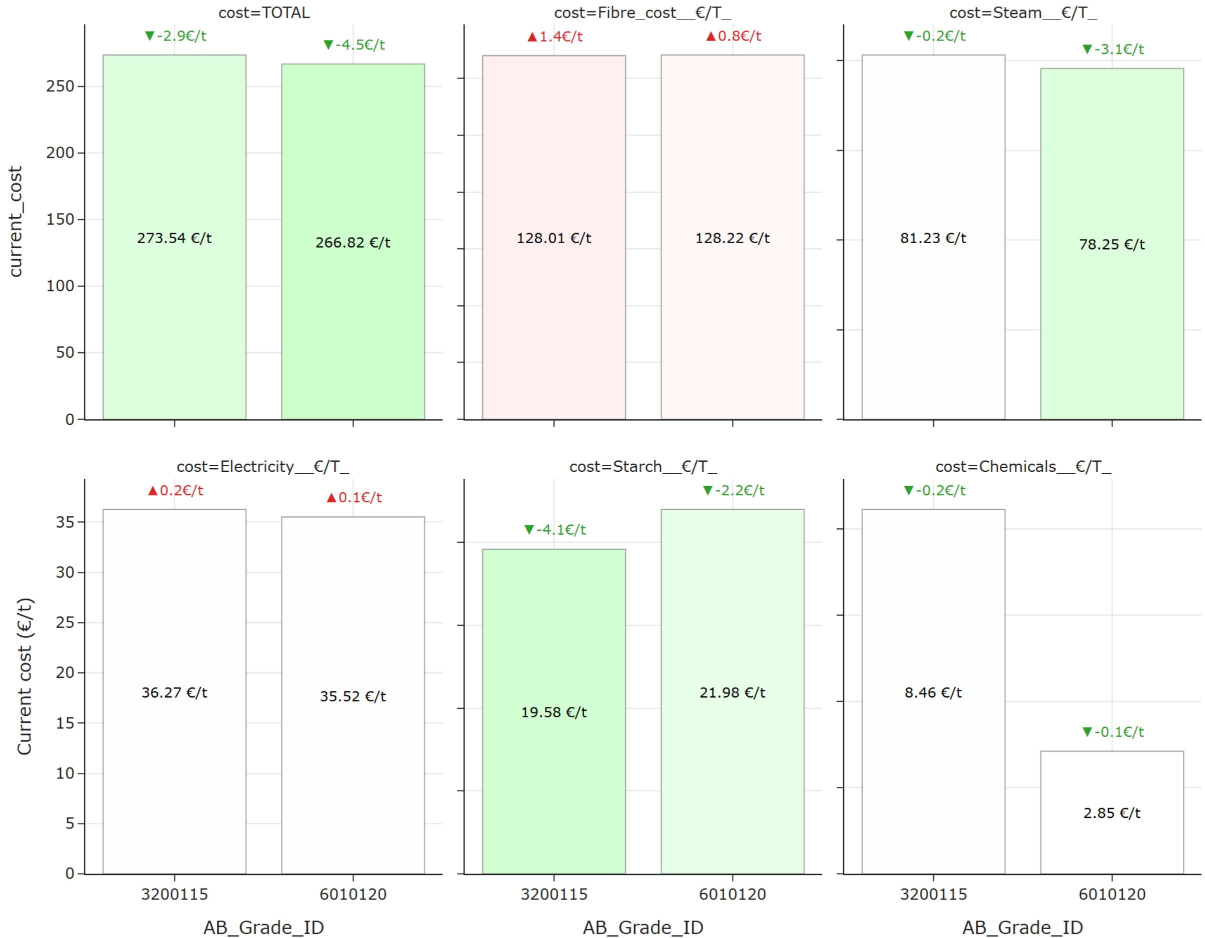
Breaking down the overall cost change, -2.18 €/t was driven by a cheaper mix of grades, and -3.83 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 2.91 €/t, resulting in 273.54 €/t.

For grade 6010120, TOTAL cost saw a decrease of 4.54 €/t, resulting in 266.82 €/t.

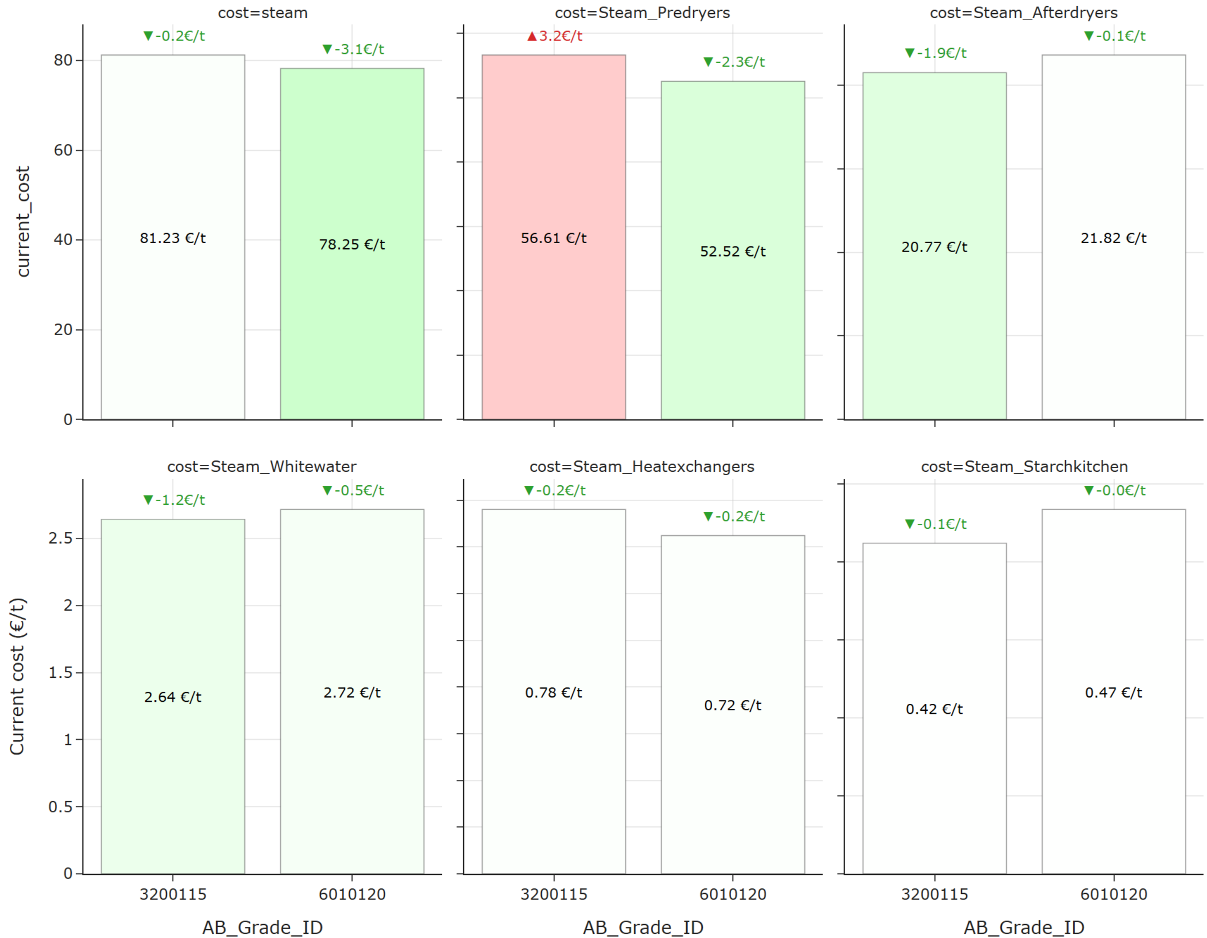
Looking across individual grades, the strongest cost decrease was observed for grade 6010120 (-4.54 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Starch__€/T** decreased on average (-3.14 €/t).
- **Steam__€/T** decreased on average (-1.67 €/t).
- **Fibre_cost__€/T** increased on average (+1.11 €/t).
- **Chemicals__€/T** decreased on average (-0.17 €/t).
- **Electricity__€/T** increased on average (+0.14 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fibre_cost__€/T** (+1.11 €/t on average), with the highest increase in grade 3200115 (+1.38 €/t) and the largest average decrease comes from **Starch__€/T** (-3.14 €/t on average), with the biggest drop in grade 3200115 (-4.06 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Afterdryers** decreased on average (-1.01 €/t).
- **Steam_Whitewater** decreased on average (-0.86 €/t).
- **Steam_Predryers** increased on average (+0.44 €/t).
- **Steam_Heatexchangers** decreased on average (-0.17 €/t).
- **Steam_Starchkitchen** decreased on average (-0.07 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Predryers** (+0.44 €/t on average), with the highest increase in grade 3200115 (+3.16 €/t) and the largest average decrease comes from **Steam_Afterdryers** (-1.01 €/t on average), with the biggest drop in grade 3200115 (-1.92 €/t). These components stand out and may require further investigation.

Average change by electricity:

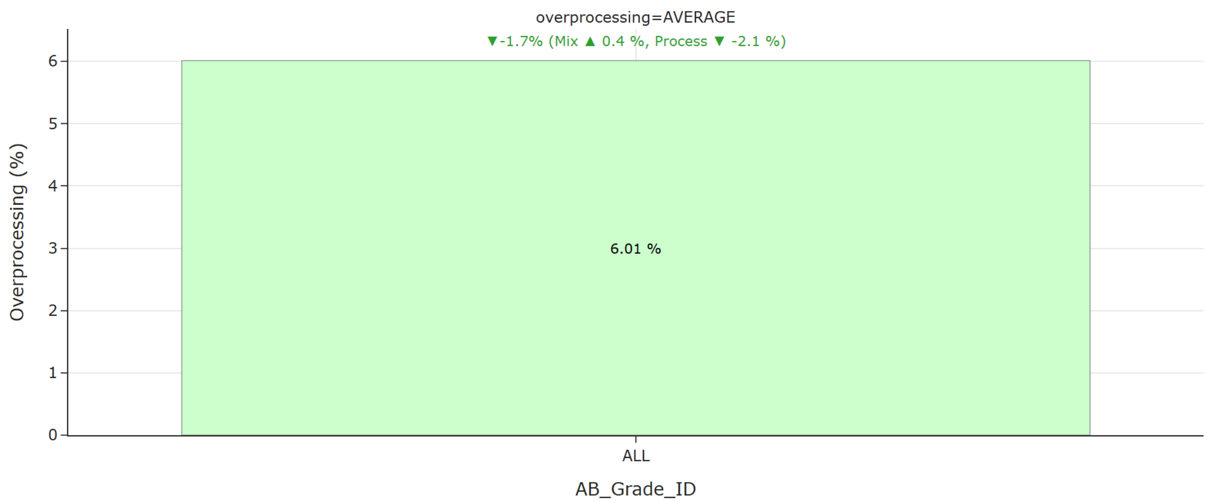
- **Electricity_Specific** increased on average (+0.44 €/t).
- **Electricity_Freqconverters** decreased on average (-0.17 €/t).
- **Electricity_Other** decreased on average (-0.17 €/t).
- **Electricity_Wiresection** increased on average (+0.11 €/t).
- **Electricity_Afterdryer** decreased on average (-0.03 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.03 €/t).
- **Electricity_Predryer** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Specific** (+0.44 €/t on average), with the highest increase in grade 6010120 (+0.48 €/t) and the largest average decrease comes from **Electricity_Freqconverters** (-0.17 €/t on average), with the biggest drop in grade 6010120 (-0.22 €/t). These components stand out and may require further investigation.

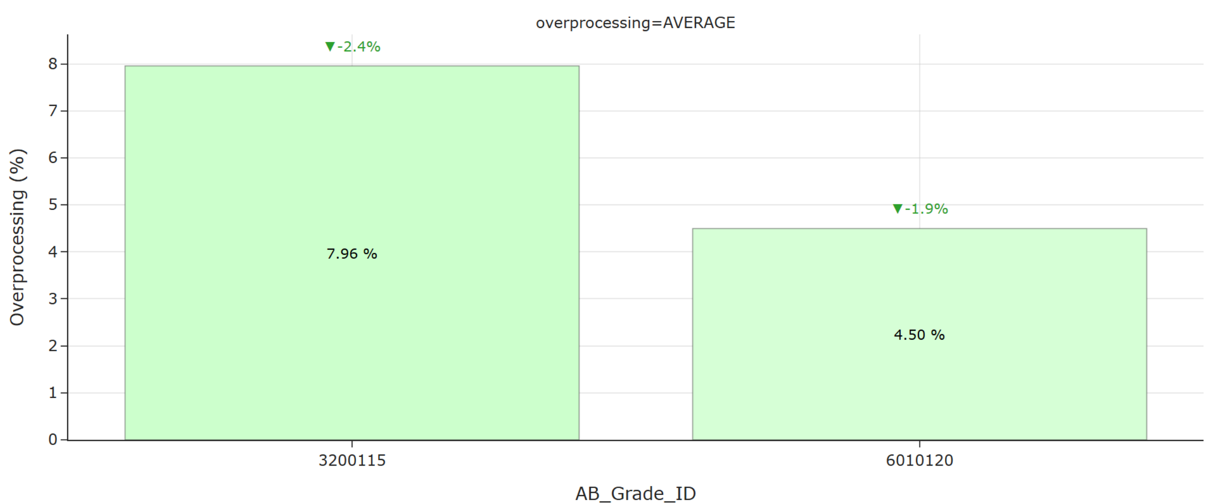
Average change by chemicals:

- **Fixative_2_€/T** decreased on average (-0.75 €/t).
- **Sizing_Agent_€/T** increased on average (+0.35 €/t).
- **Retention_Aid_€/T** increased on average (+0.10 €/t).
- **Defoamer_€/T** increased on average (+0.01 €/t).
- **Natriumhydroxide_€/T** increased on average (+0.01 €/t).
- **Bentonite_1_€/T** increased on average (+0.00 €/t).
- **Bentonite_2_€/T** increased on average (+0.00 €/t).
- **Deaerator_€/T** increased on average (+0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.35 €/t on average), with the highest increase in grade 3200115 (+0.35 €/t) and the largest average decrease comes from **Fixative_2_€/T** (-0.75 €/t on average), with the biggest drop in grade 3200115 (-0.75 €/t). These components stand out and may require further investigation.



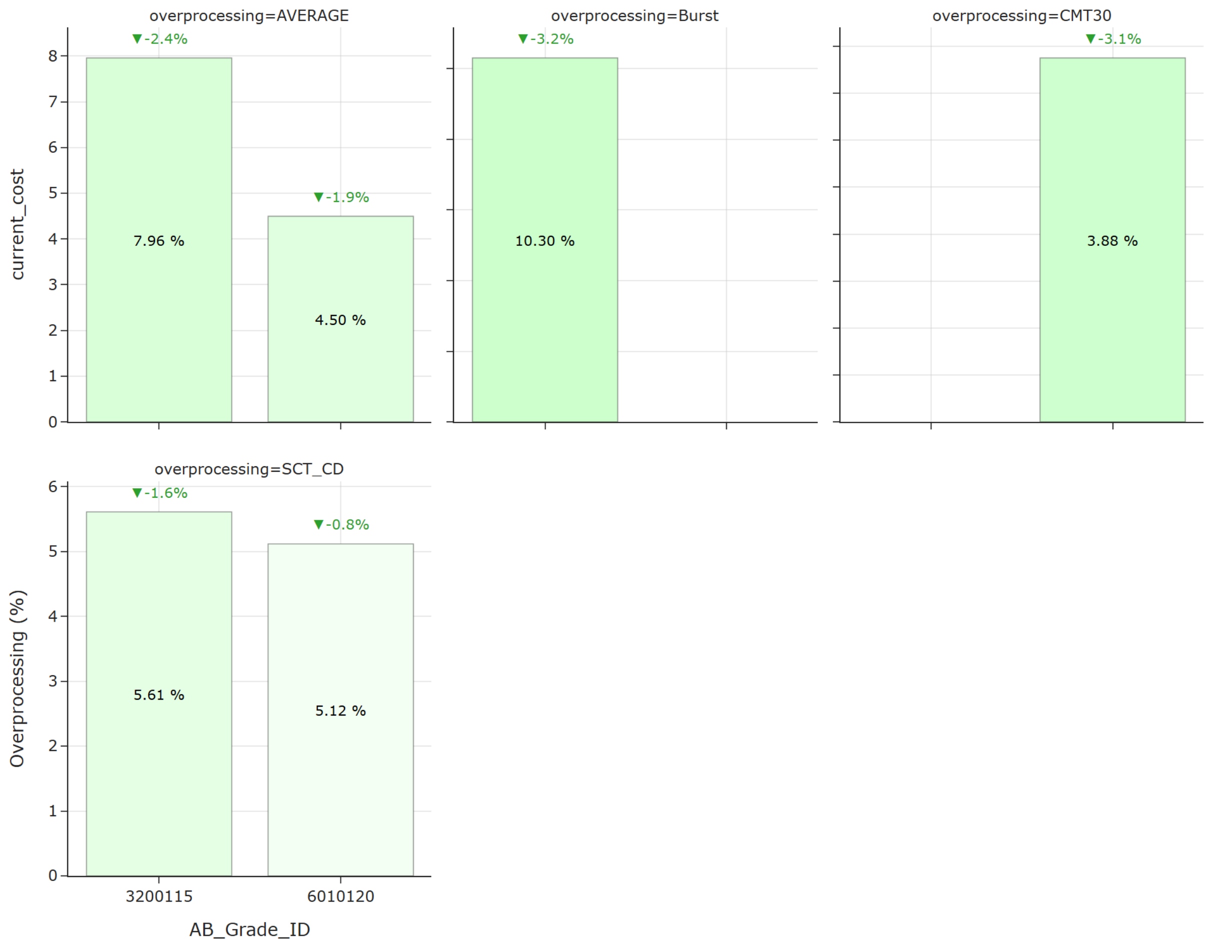
For all considered grades, TOTAL overprocessing saw a decrease of 1.72 %, resulting in 6.01 %.



For grade 3200115, TOTAL overprocessing saw a decrease of 2.41 %, resulting in 7.96 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 1.94 %, resulting in 4.50 %.

Looking across individual grades, the strongest overprocessing decrease was observed for grade 3200115 (-2.41 %). These movements stand out and may require further investigation.



Average change by component:

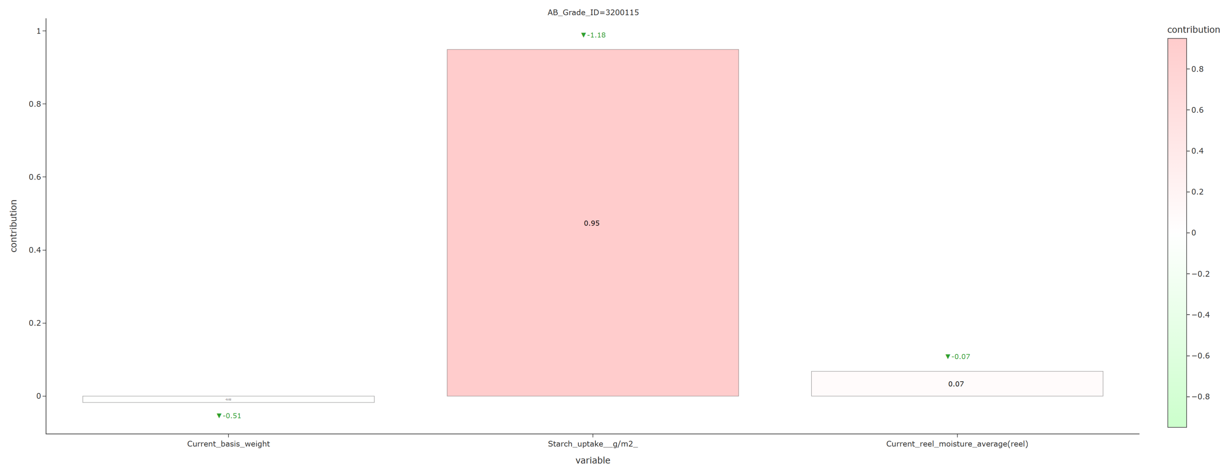
- **Burst** decreased on average (-3.20 %).
- **CMT30** decreased on average (-3.08 %).
- **SCT_CD** decreased on average (-1.21 %).

A more detailed breakdown by component shows that the largest average decrease comes from **Burst** (-3.20 % on average), with the biggest drop in grade 3200115 (-3.20 %). These components stand out and may require further investigation.

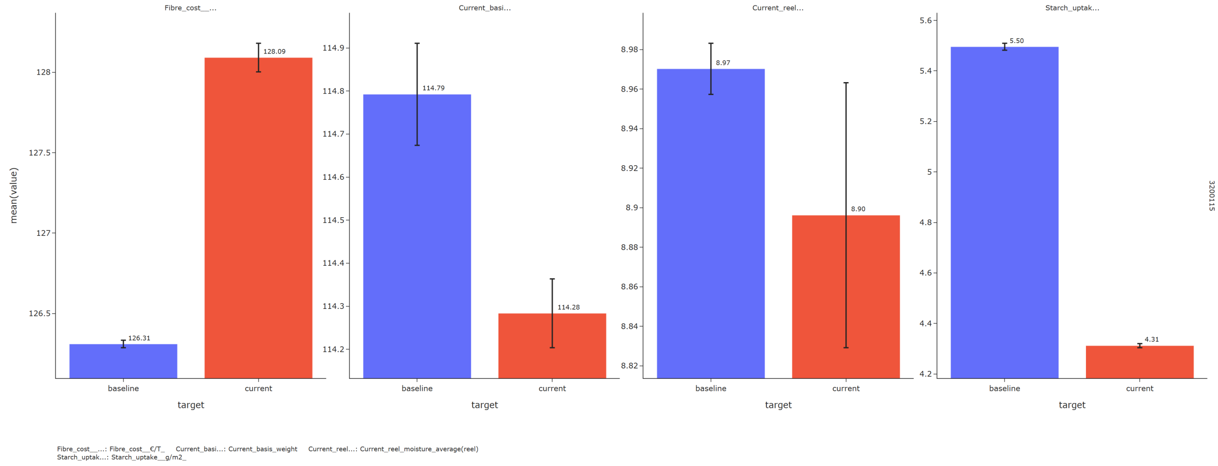
3 Breakdown by grades

3.1 3200115

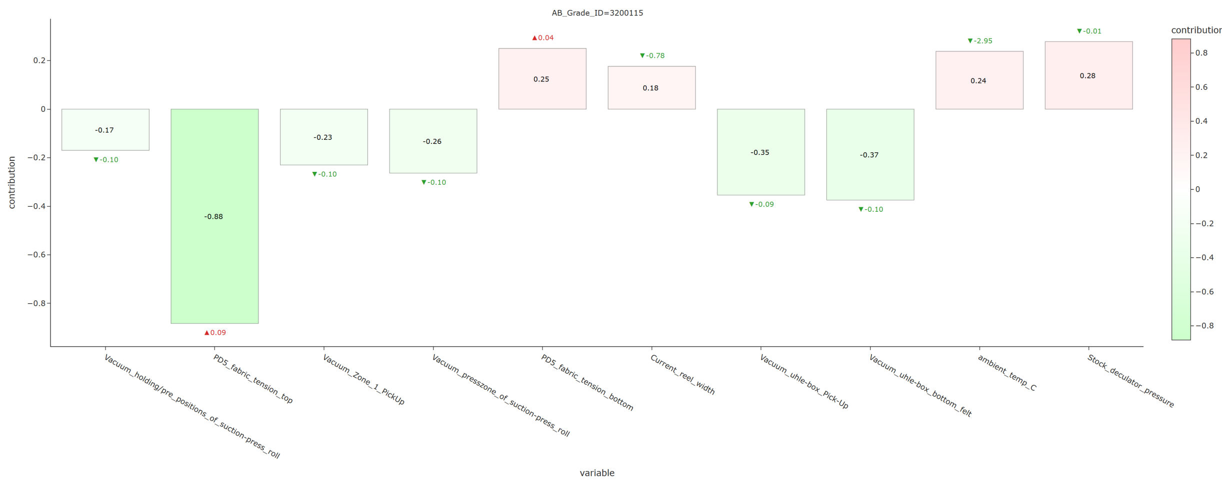
3.1.1 Fiber cost



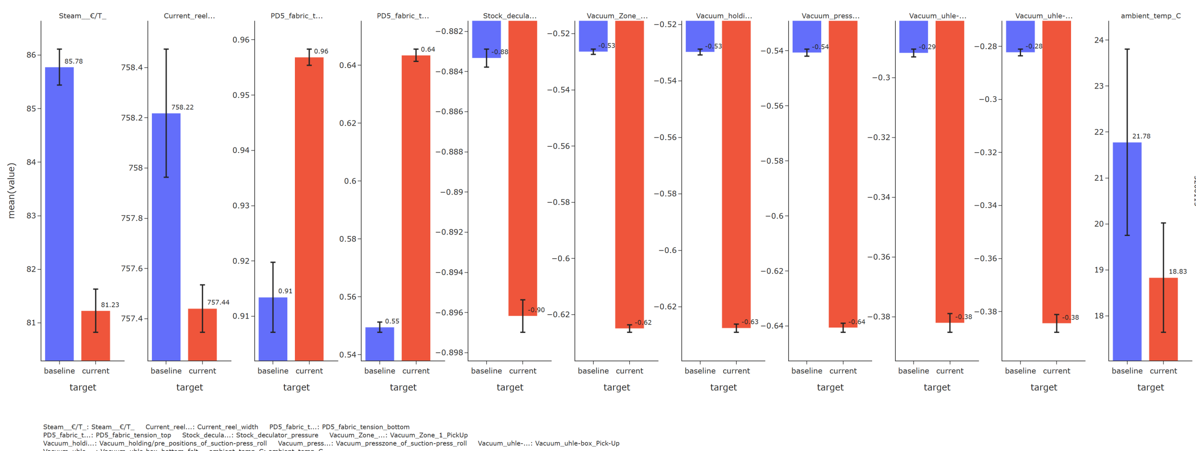
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Starch uptake g/m2 decreased (-1.18)



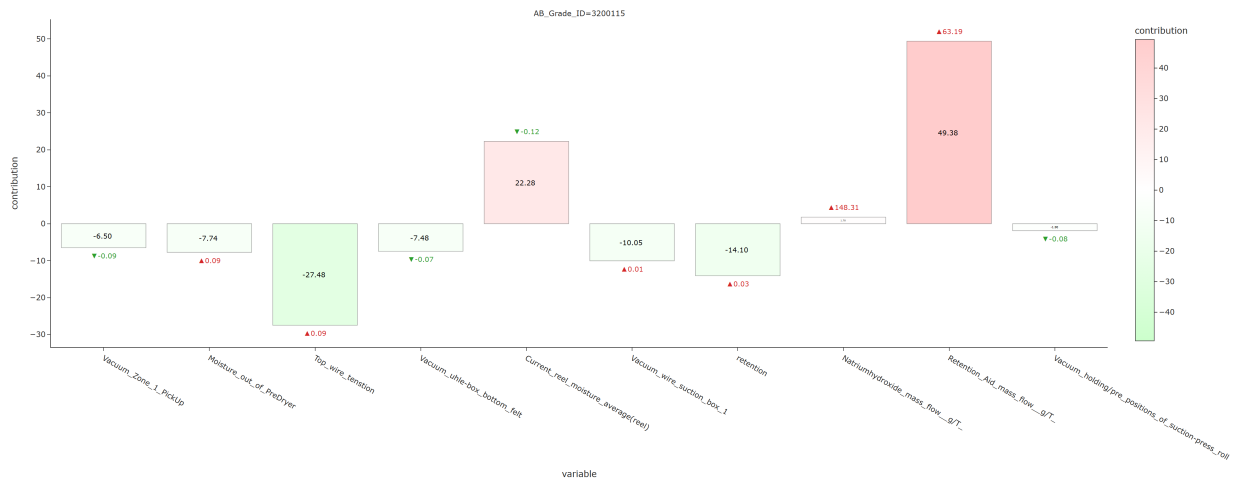
3.1.2 Steam cost



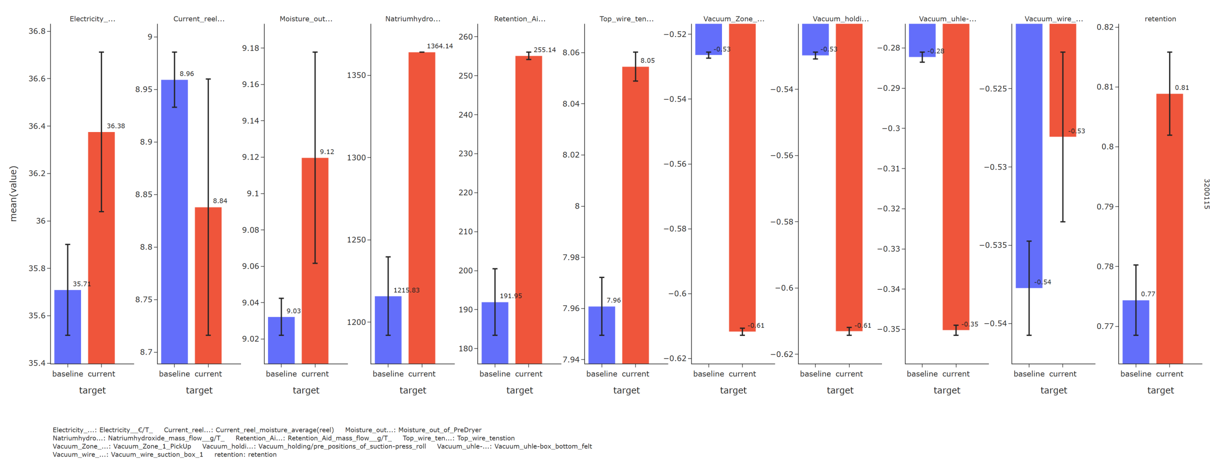
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - PD5 fabric tension top increased (+0.09) - Vacuum uhle-box bottom felt decreased (-0.10) - Vacuum uhle-box Pick-Up decreased (-0.09) - Vacuum presszone of suction-press roll decreased (-0.10) **Cost-increasing drivers**: - Stock deculator pressure decreased (-0.01)



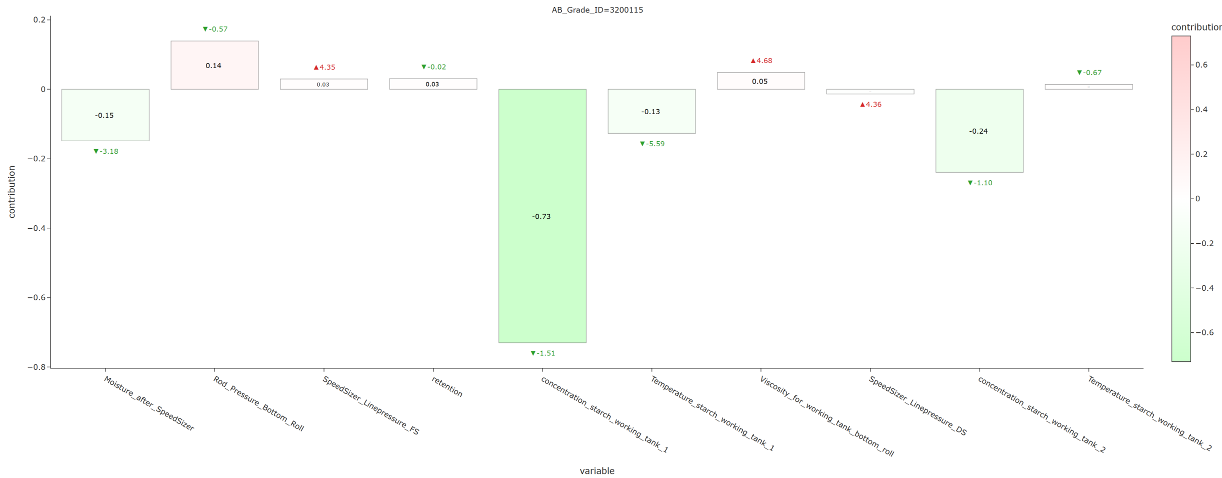
3.1.3 Electricity cost



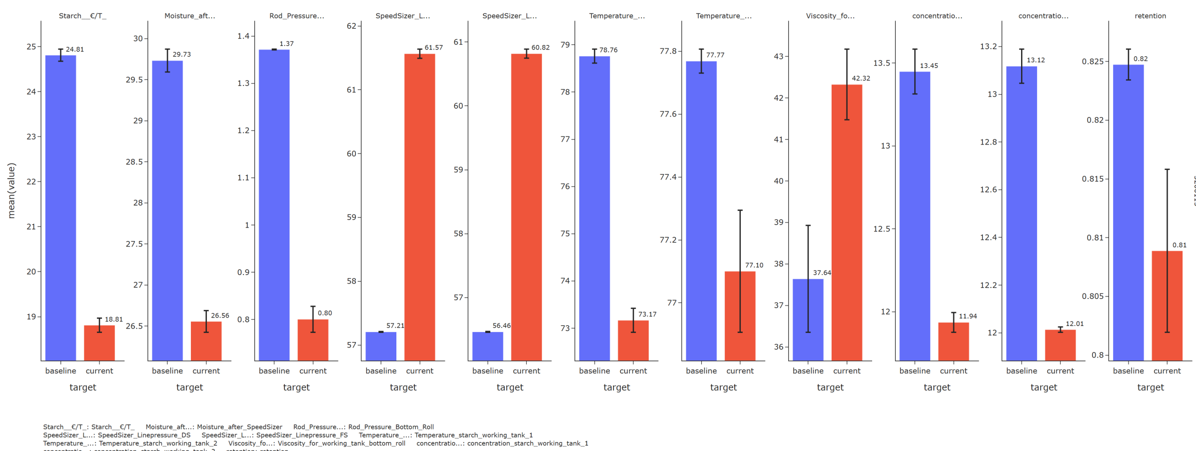
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Top wire tenstion increased (+0.09) **Cost-increasing drivers**: - Retention Aid mass flow g/T increased (+63.19) - Current reel moisture average(reel) decreased (-0.12)



3.1.4 Starch cost



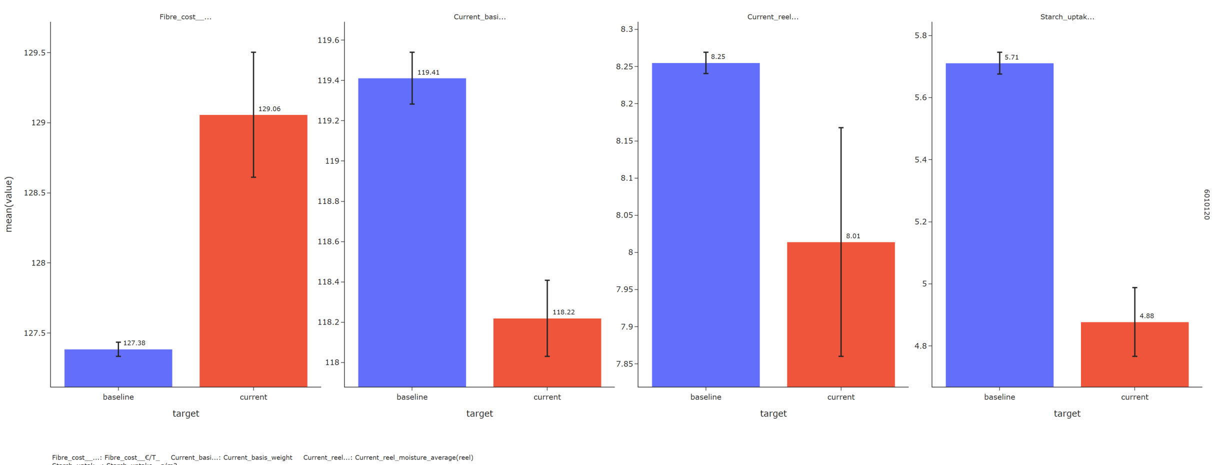
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Concentration starch working tank 1 decreased (-1.51) - Concentration starch working tank 2 decreased (-1.10) - Moisture after SpeedSizer decreased (-3.18)



3.2 6010120

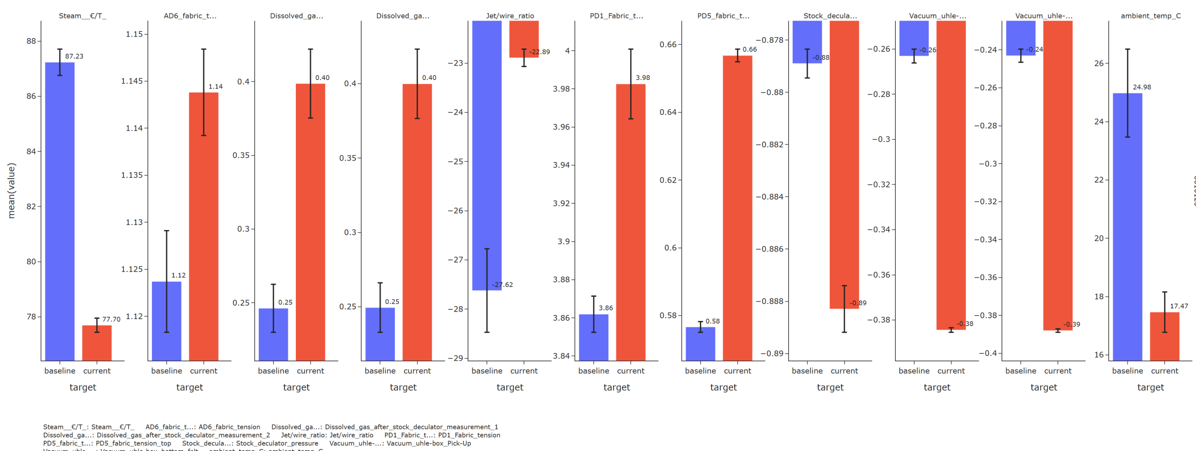
3.2.1 Fiber cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Starch uptake g/m2 decreased (-0.83)



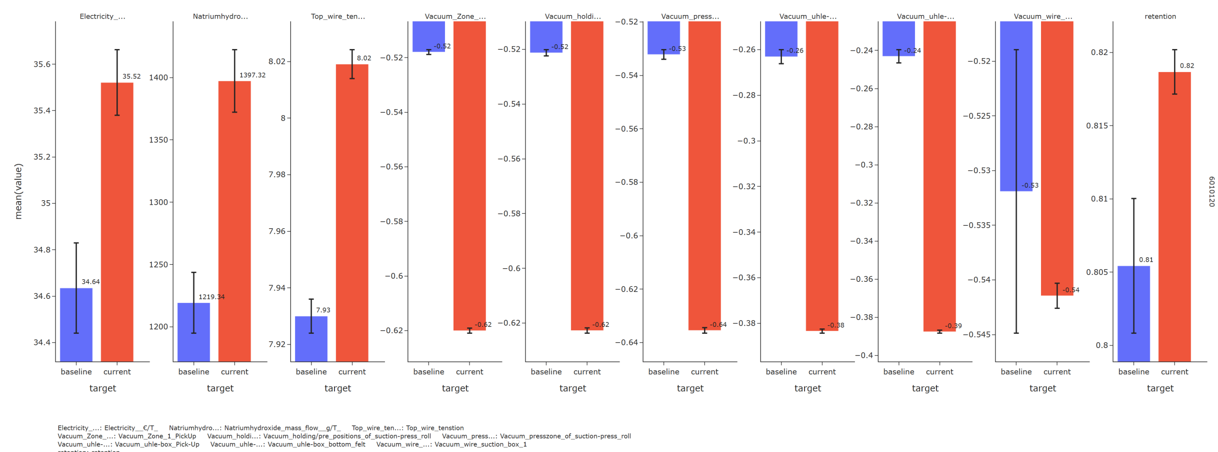
3.2.2 Steam cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - PD5 fabric tension top increased (+0.08) - Vacuum uhle-box bottom felt decreased (-0.14) - Vacuum uhle-box Pick-Up decreased (-0.12) - Jet/wire ratio increased (+4.73) **Cost-increasing drivers**: - Ambient temp C decreased (-7.51) - Stock deculator pressure decreased (-0.01)

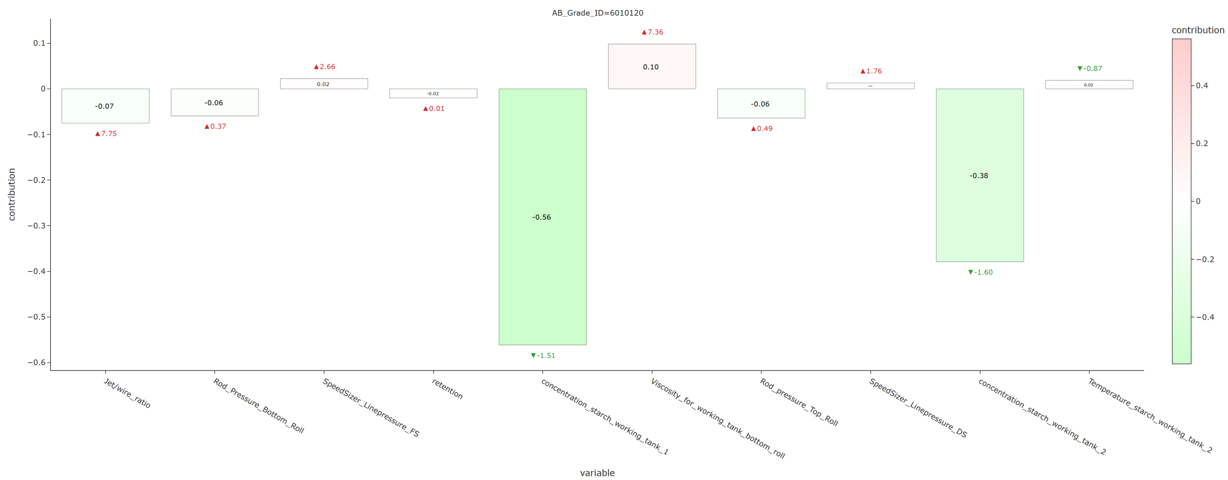


3.2.3 Electricity cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Top wire tenstion increased (+0.09) - Vacuum wire suction box 1 decreased (-0.01)



3.2.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Concentration starch working tank 1 decreased (-1.51) - Concentration starch working tank 2 decreased (-1.60) **Cost-increasing drivers**: - Viscosity for working tank bottom roll increased (+7.36)